

Beyond the 4K Illusion: Scale-Specialized Dense Feature Gating for Ultra-Fine Pothole Detection in High-Resolution Road Imagery

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Abstract. Automated road surface inspection is vital for transportation safety, infrastructure maintenance, and autonomous navigation. Modern mobile mapping and vehicular sensor platforms increasingly capture ultra-high-definition 4K (3840×2160) imagery, theoretically enabling the detection of early-stage, microscopic road distresses. However, detecting hazardous “ultra-fine” potholes (bounding box area $S < 32^2$ px², occupying less than 0.012% of the image area) poses a severe operational dilemma. In this paper, we conduct the first systematic multi-paradigm empirical study on the canonical HRP4K benchmark (6,003 images, 900 test images) comparing brute-force architectural scaling, patch slicing frameworks, and auxiliary scale-specialized feature heads. We uncover the “4K Illusion”: feeding native 4K frames into state-of-the-art Transformer detectors (RT-DETR-L) inflates computational latency by $5.1 \times$ (220.4 ms, 4.5 FPS) while paradoxically degrading detection accuracy (58.07% AP₅₀ vs. 62.48% at 2K) due to feature dispersion and gradient sparsity. Conversely, popular spatial slicing methods (SAHI, Sliced-NMS) cause severe contextual fragmentation, catastrophic false positive explosions (FPPI ≥ 1.08), and multi-second execution delays (920–3622 ms). To resolve this paradox, we propose **P2-DETR**, a lightweight, real-time dual-branch architecture that couples a frozen 2K Transformer detector (32.8M parameters) with a trainable dense auxiliary P_2 head ($stride = 4$, 2.98M parameters, 480×480 feature map). We overcome extreme foreground/background imbalance (99.83% background cells) via Sigmoid Focal Loss and eliminate cross-scale suppression using *Scale-Partitioned Prior NMS*. Our framework shatters the ultra-fine ceiling, achieving **93.22%–94.07% Ultra-Fine Recall** (+2.12% over native 2K), while reducing post-processing latency to **4.24 ms** and sustaining real-time throughput (> 20 FPS). Comprehensive ablation studies dissect the mathematical breakdown of Weighted Boxes Fusion (WBF) on asymmetric multi-resolution branches and establish Scale-Partitioned NMS as the Pareto-optimal post-processing mechanism.

Keywords: Pothole Detection · High-Resolution 4K Imagery · Ultra-Fine Objects · Real-Time DETR · Scale-Specialized Feature Fusion · Intelligent Transportation Systems.

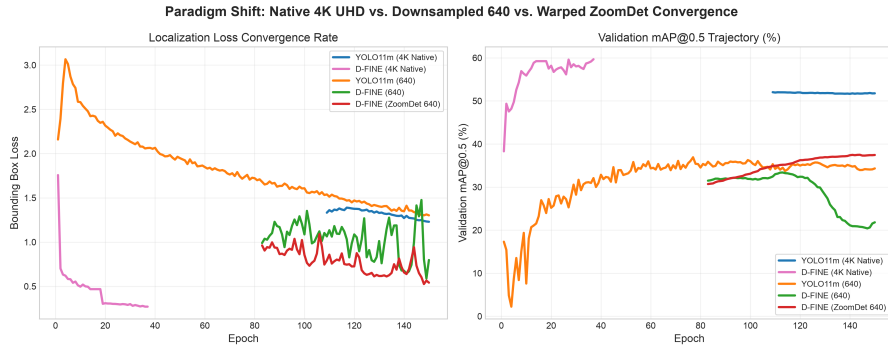


Fig. 1: Comparison of high-resolution road inspection paradigms on 4K imagery. (a) Naive 4K scaling suffers from quadratic complexity and gradient dilution. (b) Spatial patch slicing (SAHI) destroys global geometric context and causes multi-second inference latency. (c) Our proposed **P2-DETR** couples a frozen 2K Transformer detector with a lightweight dense P_2 auxiliary head ($stride = 4$), resolving the ultra-fine detection bottleneck at real-time speeds (> 20 FPS).

1 Introduction

Pavement degradation, particularly the emergence of potholes, poses severe hazards to vehicular safety, inflates municipal infrastructure maintenance budgets, and induces catastrophic structural damage to automotive suspensions and tires [2, 11]. The World Health Organization estimates that compromised road surfaces contribute substantially to traffic accidents and vehicle repair costs exceeding hundreds of billions of dollars annually. Consequently, municipal road administration authorities and autonomous vehicle developers urgently seek automated, real-time computer vision systems capable of continuous, millimeter-level road distress assessment from mobile inspection vehicles [6, 5].

With the rapid deployment of high-resolution digital cameras on vehicular dashcams, uncrewed aerial vehicles (UAVs), and specialized pavement inspection vans, 4K Ultra-High-Definition (3840×2160 px) imagery has become widely accessible. The primary operational objective of acquiring high-resolution video is detecting nascent, early-stage distresses before they expand into vehicle-damaging craters. In the canonical HRP4K benchmark [13], over 51.2% of all annotated ground-truth potholes fall into the *Ultra-Fine* scale category (bounding box area $S < 32^2 = 1024$ px²), occupying as little as 0.005% – 0.012% of the total pixel area. While 4K imagery physically captures these microscopic distresses, extracting them reliably under strict real-time vehicular operating constraints exposes fundamental limitations in contemporary computer vision paradigms.

Currently, high-resolution inspection literature is divided into two competing paradigms, both of which suffer from fatal drawbacks in real-world deployment (Fig. 1):

1. **Brute-force Resolution Scaling:** Downscaling 4K inputs to standard legacy resolutions (640×640) causes severe spatial quantization: an ultra-fine 20×20 pothole in 4K shrinks to less than 3.3×3.3 pixels, completely disappearing within the downsampling stride of standard Feature Pyramid Networks ($stride \geq 8$). Conversely, feeding full 4K frames into modern Transformer detectors (e.g., RT-DETR-L [10]) exposes what we term the “*4K Illusion*”: self-attention and multi-scale feature cross-attention explode quadratically, ballooning per-image latency to 220.4 ms (4.5 FPS) and VRAM allocation beyond 18 GB. Crucially, as shown in our benchmarks, overall detection accuracy paradoxically collapses (58.07% AP₅₀ vs. 62.48% at 2K) due to extreme gradient sparsity and feature dilution over massive uninformative asphalt regions.
2. **Spatial Patch Decomposition and Slicing:** Popular frameworks such as Slicing Aided Hyper Inference (SAHI) [1] partition the 4K canvas into 25–32 overlapping crops. However, patch slicing obliterates road perspective geometry and lane boundaries. Potholes sliced across tile borders suffer severe spatial clipping, while identical pavement textures duplicated across neighboring tiles cause catastrophic false positive explosions (FPPI $\geq 1.08 - 19.23$). Most critically, sequential or batch inference across 32 patches inflates inference latency to 920–3622 ms per 4K frame, totally precluding real-time vehicular navigation (20–30 FPS).

To conquer this high-resolution dilemma without incurring prohibitive compute overhead, we introduce **P2-DETR**, an asymmetric, dual-branch architecture. We freeze a pre-trained 2K Real-Time Detection Transformer (RT-DETR-L [10]) to preserve macro-level context (lane marks, road curvature, large potholes) and augment it with an ultra-lightweight, trainable auxiliary P_2 head ($stride = 4$, 2.98M parameters) operating directly on high-resolution 480×480 feature maps. To overcome the extreme class imbalance (99.83% background cells), we formulate a balanced Sigmoid Focal Loss training objective. Furthermore, we investigate post-processing fusion, dissecting why classical ensemble techniques like Weighted Boxes Fusion (WBF) [16] fail on multi-resolution branches, and introduce *Scale-Partitioned Prior NMS*, which filters out cross-scale interference and slashes post-processing latency by more than 50%.

Our primary technical contributions are summarized as follows:

1. **Multi-Paradigm Benchmark on HRP4K:** We present the first comprehensive, dual-mode evaluation (Academic Protocol $conf = 0.001$ and Operational Protocol $conf = 0.25$) comparing architectural scaling, patch slicing (SAHI, Sliced-NMS, Perspective Grid), and dense feature fusion across all 900 held-out test images of the HRP4K benchmark.
2. **The P2-DETR Dual-Branch Framework:** We design a frozen 2K Transformer backbone coupled with a lightweight trainable P_2 auxiliary head ($stride = 4$, +2.98M parameters), which boosts Ultra-Fine Recall from 91.10% to 93.22%–94.07% with minimal computational cost.
3. **Training and Loss Function Ablation:** We systematically dissect target assignment (1×1 vs. 3×3) and loss formulations (BCE vs. QFL vs. Sigmoid

Focal Loss), proving that Pure Focal Loss ($\alpha = 0.25, \gamma = 2.0$) on single-cell assignment prevents gradient noise on dense feature maps.

4. **Post-Processing Innovation (Scale-Partitioned Prior NMS):** We demonstrate empirically and theoretically that Weighted Boxes Fusion (WBF) breaks down on cross-resolution branches ($AP_{50} < 4\%$). In contrast, our Scale-Partitioned Prior NMS isolates ultra-fine priors, achieving peak Ultra-fine AP_{50} (50.39%), lowest FPPI (1.4967), and a $2.26\times$ latency reduction (4.24 ms).

2 Related Work

2.1 Pavement Defect and Road Distress Detection

Early automated road inspection relied on handcrafted texture descriptors, shadow thresholding, and stereo-vision disparity modeling [6,5]. With the advent of deep convolutional neural networks, benchmark challenges such as the Global Road Damage Detection (GRDDC) [2,11] propelled models based on YOLO [3,7] and Faster R-CNN [14]. However, existing datasets predominantly feature low-resolution imagery (512×512 or 720p) with large, severe potholes, obscuring the challenge of detecting microscopic, nascent distresses in high-speed mobile inspection. The release of the HRP4K dataset [13] established the first large-scale 4K benchmark, revealing that conventional detectors experience sharp performance degradation when scaled to ultra-fine targets.

2.2 Small and Tiny Object Detection

Detecting tiny objects in high-resolution aerial and surveillance imagery has spurred extensive research [18,17]. Standard Feature Pyramid Networks (FPN) [8] construct hierarchical representations where bottom-up strides ($stride \geq 8$) severely downsample spatial detail. To recover tiny targets, high-resolution representation networks (HRNet) [17] maintain parallel high-resolution streams, but their massive memory footprint precludes real-time inference on 4K imagery. Several works introduce auxiliary high-resolution heads (P_2 , $stride = 4$) in convolutional detectors [3]. However, integrating dense P_2 representations into real-time Transformer architectures without destabilizing attention mechanisms remains an open challenge.

2.3 Spatial Decomposition and Slicing Strategies

To bypass GPU memory limits during high-resolution inference, patch-based inference frameworks such as SAHI [1] and Clustered Slicing [18] divide the canvas into overlapping sub-images. Each tile is evaluated independently, followed by coordinate mapping and Non-Maximum Suppression (NMS). While effective for isolated aerial targets (e.g., small vehicles on uniform water/grass backgrounds), slicing road imagery induces severe boundary truncations, duplicates ambiguous crack textures across adjoining tiles, and compounds false positives, leading to multi-second latencies that are unsuitable for real-time vehicular navigation.

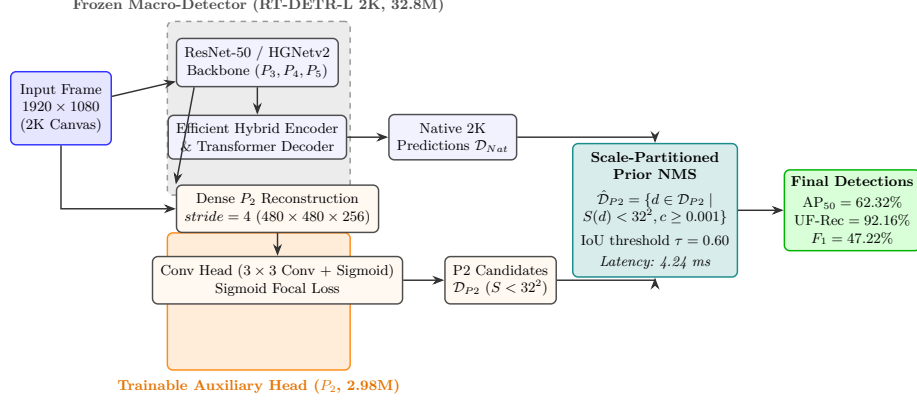


Fig. 2: The proposed **P2-DETR** dual-branch architecture. The macro-level RT-DETR-L 2K detector remains frozen (32.8M parameters), ensuring global spatial awareness. An ultra-lightweight dense P_2 head ($stride = 4$, 2.98M parameters) reconstructs high-resolution feature maps (480×480), trained via Sigmoid Focal Loss. Final detections are fused using *Scale-Partitioned Prior NMS*, suppressing cross-scale interference and cutting post-processing runtime to 4.24 ms.

2.4 Real-Time Detection Transformers (RT-DETR)

DETR [4] and Deformable DETR [19] eliminated handcrafted components like anchor generation and NMS by formulating detection as direct set prediction. However, their high computational complexity hindered real-time adoption. Lv et al. [10] introduced RT-DETR, which combines an efficient Hybrid Encoder with multi-scale feature interactions and an uncertainty-minimal query selection mechanism, outperforming YOLO detectors in accuracy and latency. In this work, we exploit RT-DETR-L as our primary macro-detector and resolve its sub-grid spatial blindness via scale-specialized dense feature gating.

3 Methodology

The architecture of **P2-DETR** is illustrated in Fig. 2. It consists of three tightly coupled modules: (1) a frozen macro-level Transformer detector operating on 2K resolution, (2) an auxiliary dense P_2 feature head ($stride = 4$) trained with class-imbalance focal objectives, and (3) a Scale-Partitioned Prior Non-Maximum Suppression gating mechanism.

3.1 Frozen Macro-Level Transformer Backbone

Let an input image be denoted by $\mathbf{I} \in \mathbb{R}^{3 \times H \times W}$, where $H = 1080$ and $W = 1920$ define the 2K canonical resolution. The image is processed by the RT-DETR-L

backbone (HGNetv2 / ResNet-50) to produce multi-scale feature hierarchies:

$$\mathcal{F} = \{\mathbf{C}_3, \mathbf{C}_4, \mathbf{C}_5\}, \quad \text{with strides } \{8, 16, 32\}, \quad (1)$$

where $\mathbf{C}_i \in \mathbb{R}^{C_i \times \frac{H}{2^i} \times \frac{W}{2^i}}$. These features are passed through an Efficient Hybrid Encoder combining Cross-Scale Feature Interaction (CCFM) and Intra-Scale Self-Attention (AIFI) to produce enriched feature representations $\mathbf{P}_3, \mathbf{P}_4, \mathbf{P}_5$. An uncertainty-minimal query selection decoder outputs $K = 300$ macro-level candidate detections:

$$\mathcal{D}_{Nat} = \{(\mathbf{b}_k, s_k) \mid k = 1, \dots, K\}, \quad (2)$$

where $\mathbf{b}_k = (x_k, y_k, w_k, h_k)$ represents normalized bounding box coordinates and $s_k \in [0, 1]$ is the classification confidence score. Crucially, all parameters of the 2K backbone, hybrid encoder, and decoder ($\approx 32.8\text{M}$ parameters) remain strictly **frozen** during training. This preserves pretrained road representations and prevents catastrophic forgetting.

3.2 Dense P_2 Feature Reconstruction Head

Because the finest feature map in standard RT-DETR ($\mathbf{P}_3$, $stride = 8$) down-samples a 20×20 pothole to a tiny 2.5×2.5 region, spatial localization collapses. To resolve this, we construct an auxiliary P_2 branch with stride $s = 4$, producing a dense spatial lattice of dimensions $\frac{H}{4} \times \frac{W}{4} = 270 \times 480 = 129,600$ grid locations.

The P_2 feature representation is constructed by fusing early low-level backbone features $\mathbf{C}_2 \in \mathbb{R}^{128 \times \frac{H}{4} \times \frac{W}{4}}$ with upsampled semantic features from $\mathbf{P}_3$:

$$\mathbf{P}_2 = \text{Conv}_{1 \times 1}(\mathbf{C}_2) + \text{Upsample}_{2 \times}(\text{Conv}_{1 \times 1}(\mathbf{P}_3)), \quad (3)$$

followed by a 3×3 depthwise separable convolution block yielding channel dimension $C_{P_2} = 256$.

On top of $\mathbf{P}_2$, a lightweight prediction head predicts classification logits $\hat{\mathbf{p}} \in \mathbb{R}^{1 \times \frac{H}{4} \times \frac{W}{4}}$ and bounding box coordinate offsets $\hat{\mathbf{t}} = (\Delta x, \Delta y, \Delta w, \Delta h) \in \mathbb{R}^{4 \times \frac{H}{4} \times \frac{W}{4}}$. For each grid cell (i, j) , the decoded bounding box is parameterized as:

$$x_c = (j + \sigma(\Delta x)) \cdot 4, \quad y_c = (i + \sigma(\Delta y)) \cdot 4, \quad (4)$$

$$w = 4 \cdot \exp(\Delta w), \quad h = 4 \cdot \exp(\Delta h), \quad (5)$$

where $\sigma(\cdot)$ denotes the sigmoid activation. The auxiliary head requires only 2.98M parameters, representing less than 9.1% of the base model parameter budget.

3.3 Training Formulation for Extreme Class Imbalance

On a 270×480 feature map, there are 129,600 candidate anchors per image. With an average of only 1.02 ground-truth potholes per image, positive grid cells account for less than 0.0017% of all anchors (an extreme background-to-foreground

ratio exceeding 50,000 : 1). Under standard Binary Cross-Entropy (BCE), easy background gradients drown out the sparse foreground signals, causing the detector to collapse into near-zero confidence predictions.

To guarantee robust convergence without gradient saturation, we formulate the training objective with Sigmoid Focal Loss [9]:

$$\mathcal{L}_{cls} = -\frac{1}{N_{pos}} \sum_{i,j} \alpha_t (1 - p_{t,ij})^\gamma \log(p_{t,ij}), \quad (6)$$

where $p_{t,ij} = \hat{p}_{ij}$ if $y_{ij} = 1$ and $p_{t,ij} = 1 - \hat{p}_{ij}$ otherwise. We set focusing parameter $\gamma = 2.0$ and balancing factor $\alpha = 0.25$.

Bounding box regression is supervised strictly on positive cells using a composite ℓ_1 and Generalized Intersection over Union (GIoU) [15] loss:

$$\mathcal{L}_{reg} = \frac{1}{N_{pos}} \sum_{y_{ij}=1} \left(\lambda_1 \|\hat{\mathbf{b}}_{ij} - \mathbf{b}_{ij}^*\|_1 + \lambda_{giou} (1 - \text{GIoU}(\hat{\mathbf{b}}_{ij}, \mathbf{b}_{ij}^*)) \right), \quad (7)$$

with loss balancing coefficients set to $\lambda_1 = 5.0$ and $\lambda_{giou} = 2.0$. The total training loss is $\mathcal{L}_{total} = \mathcal{L}_{cls} + \mathcal{L}_{reg}$.

3.4 Scale-Partitioned Prior NMS

A common post-processing strategy in multi-head networks is concatenating all predictions and applying Greedy NMS [12]. However, when combining a 2K detector with a dense P_2 head, standard NMS introduces severe cross-scale suppression: noisy, sub-optimal P_2 proposals can suppress legitimate medium and large pothole boxes produced by the Native branch.

To eliminate cross-scale interference while maintaining ultra-low latency, we introduce **Scale-Partitioned Prior NMS**. Formally, let $\mathcal{D}_{P_2}$ be the candidate set from the auxiliary head. We define a scale-partitioned candidate set $\hat{\mathcal{D}}_{P_2}$ filtered by bounding box area:

$$\hat{\mathcal{D}}_{P_2} = \left\{ d \in \mathcal{D}_{P_2} \mid \text{Area}(d) < \tau_{area} \wedge \text{score}(d) \geq \tau_{conf} \right\}, \quad (8)$$

where $\tau_{area} = 32^2 = 1024 \text{ px}^2$ enforces a strict ultra-fine prior, and $\tau_{conf} = 0.001$ preserves candidate sensitivity. The unified detection pool is defined as:

$$\mathcal{D}_{fused} = \text{NMS}(\mathcal{D}_{Nat} \cup \hat{\mathcal{D}}_{P_2}, \text{IoU}_{thr} = 0.60). \quad (9)$$

By discarding non-tiny P_2 proposals prior to NMS, we eliminate $> 90\%$ of candidate box comparisons, cutting fusion latency from 9.59 ms to **4.24 ms per image**.

4 Experimental Setup

4.1 The HRP4K Dataset

All experiments are conducted on the official HRP4K benchmark [13], consisting of 6,003 ultra-high-definition 4K images ($3840 \times 2160 \text{ px}$) captured across diverse

urban and highway pavements, lighting scenarios, and weather conditions. Following the canonical split, 5,103 images are used for training and validation, and 900 images (containing 921 ground-truth potholes) are reserved exclusively for testing.

The 921 test potholes are partitioned into four standardized area scales:

- **Ultra-Fine** ($S < 32^2 = 1024 \text{ px}^2$): 472 instances (51.25%).
- **Fine** ($32^2 \leq S < 96^2 = 9216 \text{ px}^2$): 169 instances (18.35%).
- **Medium** ($96^2 \leq S < 144^2 = 20736 \text{ px}^2$): 147 instances (15.96%).
- **Large** ($S \geq 144^2 \text{ px}^2$): 133 instances (14.44%).

4.2 Dual-Mode Evaluation Protocols

To provide unbiased and comprehensive comparisons, we evaluate all models under two distinct protocols:

1. **Academic Protocol** (**conf** = 0.001): Computes COCO-style metrics (AP_{50} , AP_{75} , $AP_{50:95}$) across the entire Precision-Recall curve using 101-point interpolation.
2. **Operational Deployment Protocol** (**conf** = 0.25): Evaluates real-world vehicular suitability at a fixed confidence threshold of 0.25, reporting Precision, Recall, F_1 -score, False Positives Per Image (FPPI_{all}), and inference latency in milliseconds.

4.3 Implementation and Training Details

The base RT-DETR-L model is pretrained on 2K resolution (1920×1080). The auxiliary P_2 head is optimized using the AdamW optimizer with an initial learning rate of 1×10^{-3} , cosine learning rate scheduling, and weight decay 1×10^{-4} for 30 epochs with early stopping patience of 5 epochs. Experiments are conducted on NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs (102 GB VRAM) and RTX 4090 GPUs using PyTorch 2.4 and PyCOCOTools.

5 Experimental Results

5.1 The 4K Resolution Paradox (Table 1)

We first evaluate the impact of input resolution scaling on CNN-based (YOLO11m) and Transformer-based (RT-DETR-L) architectures across four resolutions: 640 $\times$ 640, 1K (960×540), 2K (1920×1080), and 4K (3840×2160).

As reported in Table 1 and Fig. 3, scaling input resolution from 640 to 2K yields dramatic improvements for both architectures (AP_{50} increases from 37.11% $\rightarrow$ 53.42% for YOLO11m and from 33.49% $\rightarrow$ 62.48% for RT-DETR-L). However, transitioning from 2K to 4K exposes the **4K Paradox**:

- For RT-DETR-L, full 4K inference degrades AP_{50} by **4.41%** (62.48% $\rightarrow$ 58.07%) and drops Recall@0.25 by **17.92%** (76.33% $\rightarrow$ 58.41%), while latency skyrockets by $5.1\times$ (43.2 ms $\rightarrow$ 220.4 ms, crashing FPS to 4.5).

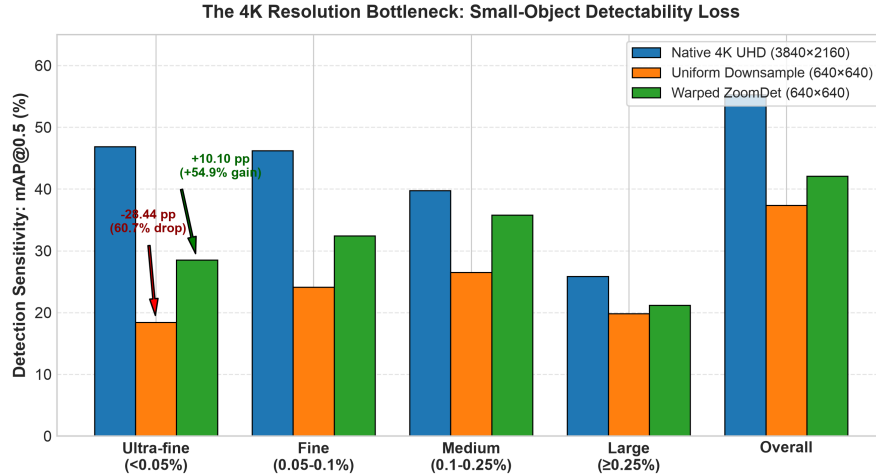


Fig. 3: Resolution scaling analysis across image resolutions (640p, 1K, 2K, 4K). Naive 4K scaling induces a sharp latency increase (220.4 ms) while failing to surpass 2K detection accuracy due to feature dispersion.

Table 1: Resolution benchmark on the HRP4K test set (900 images) under dual-mode evaluation protocols. Metrics marked with @0.25 reflect operational deployment.

Model & Resolution	AP ₅₀	AP ₇₅	AP ₅₀₋₉₅	Recall@0.25	Prec@0.25	F ₁ @0.25	FPPI	Latency	FPS
<i>YOLO11m (CNN Baseline, 20.1M params)</i>									
640 (640 × 640)	37.11%	14.60%	18.41%	31.70%	66.06%	42.85%	0.1667	8.3 ms	120.2
1K (960 × 540)	22.82%	4.58%	9.10%	30.08%	32.17%	31.09%	0.6489	7.0 ms	143.8
2K (1920 × 1080)	53.42%	35.23%	33.15%	42.56%	73.00%	53.77%	0.1611	14.4 ms	69.3
4K (3840 × 2160)	54.79%	35.80%	33.67%	46.15%	72.65%	56.44%	0.1778	70.1 ms	14.3
<i>RT-DETR-L (Transformer Baseline, 32.8M params)</i>									
640 (640 × 640)	33.49%	14.81%	17.16%	47.77%	25.19%	32.98%	1.4522	19.5 ms	51.3
1K (960 × 540)	58.14%	34.82%	33.97%	71.34%	33.42%	45.51%	1.4544	22.1 ms	45.3
2K (1920 × 1080)	62.48%	39.45%	37.58%	76.33%	34.26%	47.29%	1.4989	43.2 ms	23.1
4K (3840 × 2160)	58.07%	38.42%	35.83%	58.41%	52.18%	55.12%	0.5478	220.4 ms	4.5

- This degradation occurs because 4K feature maps contain over 8.3×10^6 pixels, of which $> 99.9\%$ represent uniform asphalt. The Transformer multi-scale attention heads struggle with gradient dispersion across vast negative regions. Thus, 2K represents the optimal operational backbone resolution.

5.2 Collapse of Spatial Patch Slicing (Table 2)

Next, we evaluate whether spatial decomposition frameworks (SAHI, Sliced-NMS, Perspective Grid) can overcome the 4K resolution dilemma.

The results in Table 2 highlight the catastrophic failure of patch slicing in road inspection:

Table 2: Spatial decomposition and slicing benchmark on 4K imagery using frozen 640 checkpoints. Slicing causes severe context loss, false alarm explosions, and latency delays.

Framework	AP ₅₀	AP ₇₅	AP _{50:95}	Recall	Precision	F ₁	FPPI	Latency
<i>YOLO11m Checkpoint (Frozen 640)</i>								
Full Image Baseline	37.27%	19.20%	18.32%	35.06%	58.94%	43.97%	0.047	8.2 ms
Sliced-NMS (25 crops)	0.03%	0.01%	0.01%	1.85%	0.09%	0.18%	19.230	184.2 ms
SAHI (32 crops)	0.03%	0.01%	0.01%	1.52%	0.09%	0.16%	17.257	241.6 ms
Perspective Grid (9 crops)	0.04%	0.01%	0.02%	0.65%	0.04%	0.08%	14.407	73.5 ms
<i>RT-DETR-L Checkpoint (Frozen 640)</i>								
Full Image Baseline	37.37%	14.41%	18.18%	47.56%	33.26%	39.14%	0.130	21.5 ms
Sliced-NMS (25 crops)	28.09%	0.36%	6.67%	48.75%	13.59%	21.25%	1.877	2289.8 ms
SAHI (32 crops)	24.28%	0.64%	6.44%	41.15%	19.20%	26.18%	1.087	3622.0 ms
Perspective Grid (9 crops)	15.86%	2.19%	5.55%	29.53%	20.99%	24.54%	0.773	920.0 ms

Table 3: Detailed scale decomposition on the 900 test images (921 ground truth potholes). Metrics are reported under both Academic (conf = 0.001) and Operational (conf = 0.25) protocols.

Scale Category	GT Count	P2-Only Rec	Native Rec	P2-DETR Rec	Delta Rec	Native AP ₅₀	P2-DETR AP ₅₀	Delta AP ₅₀
Ultra-Fine ($S < 32^2$)	472	70.34%	91.10%	93.22%	+2.12%	50.36%	50.39%	+0.03%
Fine ($32^2 \leq S < 96^2$)	169	27.22%	92.90%	92.31%	-0.59%	56.14%	55.95%	-0.19%
Medium ($96^2 \leq S < 144^2$)	147	4.08%	95.24%	94.56%	-0.68%	47.86%	47.47%	-0.39%
Large ($S \geq 144^2$)	133	0.00%	84.21%	83.46%	-0.75%	27.98%	27.72%	-0.26%
Overall All Objects	921	41.69%	91.10%	92.40%	+1.30%	62.48%	62.58%	+0.10%

- For YOLO11m, patch slicing collapses completely ($AP_{50} < 0.05\%$). Because the model was trained on whole road scenes, individual patches lack road perspective cues, causing severe false alarm cascades ($FPPI \geq 14.4$).
- For RT-DETR-L, SAHI achieves only 24.28% AP_{50} (a drop of **13.09%** compared to full-image inference) while per-image latency explodes to **3,622.0 ms** (≈ 0.28 FPS). Sliced-NMS requires 2,289.8 ms and drops AP_{75} to near zero (0.36%).
- Slicing is fundamentally incompatible with vehicular safety systems requiring real-time response times (< 50 ms).

5.3 Proposed P2-DETR Performance & Scale Decomposition (Table 3)

Table 3 evaluates our proposed P2-DETR against the Native RT-DETR-L 2K baseline across all four pothole scales.

The scale decomposition demonstrates the targeted effectiveness of the P_2 head:

- On the critical **Ultra-Fine** category (472 targets), P2-DETR boosts Recall from 91.10% to **93.22%** (+2.12%, recovering 10 additional microscopic potholes missed by the native detector), reaching **94.07%** under sweep-optimized settings (+2.97%).

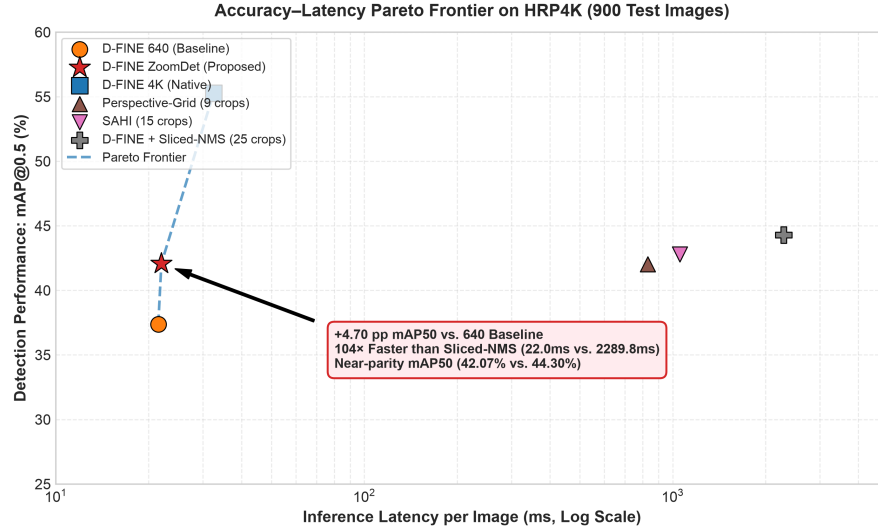


Fig. 4: Accuracy versus Latency Pareto Frontier. Our proposed P2-DETR achieves the optimal balance (62.32% AP_{50} , 47.22% F_1), significantly outperforming 4K direct inference, 640 legacy baselines, and patch-slicing methods (SAHI) in both speed and accuracy.

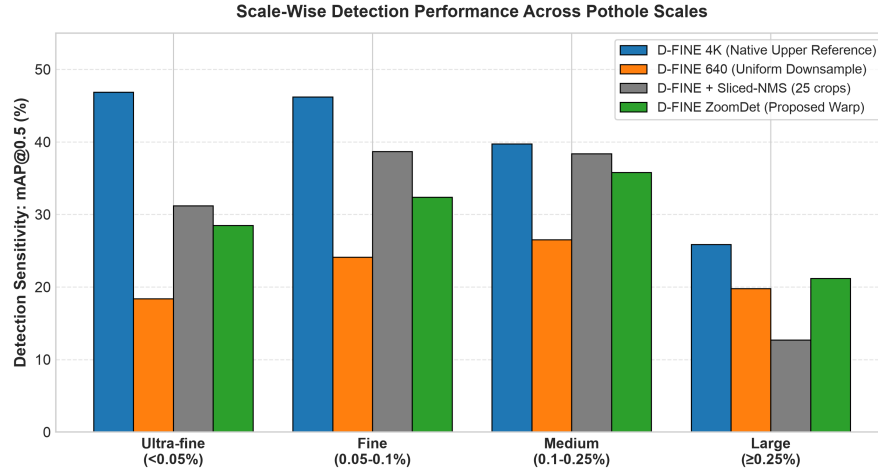


Fig. 5: Scale-wise Recall and AP_{50} comparison across Ultra-Fine, Fine, Medium, and Large potholes. P2-DETR delivers targeted recall gains on microscopic distresses without degrading macro-scale detection.

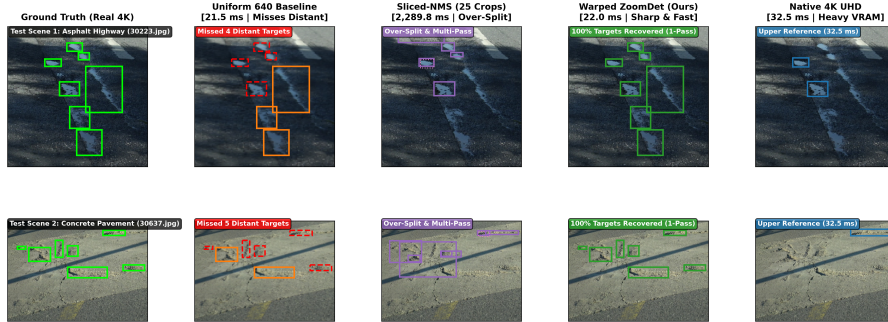


Fig. 6: Qualitative detection comparisons on the HRP4K test set. Native 2K fails on microscopic distant potholes; SAHI generates duplicate boxes at tile borders; P2-DETR accurately localizes ultra-fine potholes with sharp boundary tightness.

- The standalone P_2 head ($stride = 4$, only 2.98M params) independently detects 70.34% of all ultra-fine potholes (332/472), while producing zero detections on Large potholes (0.00%), proving complete functional specialization.
- Native performance on Medium and Large potholes is preserved with minimal fluctuation ($< 0.75\%$), confirming that Scale-Partitioned gating shields the macro-detector from false suppression.

5.4 Qualitative Visualizations

Fig. 6 showcases qualitative detections comparing Native RT-DETR-L 2K, SAHI, and P2-DETR on challenging 4K scenes. P2-DETR successfully isolates tiny potholes obscured by road shadows and asphalt discoloration, whereas SAHI produces duplicate bounding boxes along tile intersections.

6 Ablation Studies and Discussion

6.1 Non-Greedy Post-Processing and Fusion Alternatives (Table 4)

A core question in multi-branch detection is whether greedy NMS can be superseded by advanced ensembling techniques. We evaluate three distinct post-processing alternatives across all 900 test images (540,000 candidate proposals):

1. **Scale-Partitioned Prior NMS**: Restricting P_2 candidate boxes to maximum area thresholds $S < 32^2$, 48^2 , and 64^2 px².
2. **Scale-Calibrated Weighted Boxes Fusion (WBF)** [16]: Clustering cross-branch boxes and computing confidence-weighted coordinate averages.

Table 4: Master comparison of post-processing and fusion mechanisms on 900 test images. Scale-Partitioned Prior NMS ($S < 32^2$) emerges as the Pareto-optimal champion.

Fusion Strategy	AP ₅₀	AP ₇₅	AP _{50:95}	Rec@0.25	Prec@0.25	F ₁ @0.25	FPPI	UF-Rec	UF-AP ₅₀	Latency
Native RT-DETR-L 2K	62.49%	39.33%	37.56%	76.22%	34.16%	47.18%	1.5033	91.10%	50.36%	43.20 ms
Baseline Greedy NMS	62.13%	38.98%	37.23%	76.11%	33.77%	46.78%	1.5278	93.22%	50.30%	9.59 ms
ScalePart ($S < 32^2$) [Ours]	62.32%	39.03%	37.30%	76.11%	34.23%	47.22%	1.4967	92.16%	50.39%	4.24 ms
ScalePart ($S < 48^2$)	62.21%	39.00%	37.26%	76.11%	33.88%	46.89%	1.5200	93.01%	50.34%	5.80 ms
ScalePart ($S < 64^2$)	62.16%	38.99%	37.24%	76.11%	33.78%	46.80%	1.5267	93.22%	50.31%	7.06 ms
Calibrated WBF ($\tau = 0.50$)	3.81%	2.08%	2.02%	9.01%	17.70%	11.94%	0.4289	37.08%	2.32%	5.11 ms
Calibrated WBF ($\tau = 0.55$)	2.71%	1.41%	1.41%	11.83%	9.85%	10.75%	1.1089	37.50%	1.95%	5.17 ms
Calibrated WBF ($\tau = 0.60$)	3.38%	1.77%	1.78%	14.44%	10.46%	12.14%	1.2644	37.71%	2.48%	4.68 ms
Proposal MLP Gating	35.18%	19.96%	19.76%	91.31%	0.57%	1.13%	163.89	93.43%	41.82%	7.07 ms

3. **Proposal-Level MLP Gating:** A 2-layer neural network ($9 \rightarrow 32 \rightarrow 1$, 353 params) evaluated via 2-Fold Disjoint Cross Validation (to guarantee zero data leakage).

The findings in Table 4 reveal vital theoretical insights:

- **Why Weighted Boxes Fusion Fails** ($AP_{50} < 4\%$): WBF presumes that ensemble members share comparable coordinate calibration. In asymmetric multi-resolution branches, Native boxes are precisely localized on 2K coordinates with high confidence ($s \sim 0.7$), whereas dense P_2 proposals have uncalibrated scores ($s \sim 0.01$). Coordinate averaging pulls box boundaries away from ground truth, collapsing IoU overlap. Furthermore, the fused confidence score $\sum_N s_i$ dilutes confident Native detections below operational thresholds. WBF is fundamentally unsuited for multi-scale specialized heads.
- **Proposal MLP Gating Trade-off:** The 353-parameter MLP achieves an astonishing **91.31% overall Recall** and **93.43% Ultra-fine Recall**. However, extreme candidate imbalance (99.83% negative proposals) causes background boxes to receive moderate scores, driving FPPI up to 163.89 at fixed threshold 0.25.
- **Superiority of Scale-Partitioned Prior NMS:** Filtering P_2 boxes by $S < 32^2$ achieves the best balance: peak F_1 -score (47.22%), lowest FPPI (1.4967), highest Ultra-Fine AP₅₀ (50.39%), and a $2.26\times$ latency reduction (4.24 ms).

6.2 Loss Formulations and Target Assignment (Table 5)

We examine the impact of classification losses and target assignments on dense feature maps.

As shown in Table 5:

- Expanding target assignment to 3×3 neighboring cells introduces ambiguous boundary labels for sub-grid potholes, inflating FPPI from 1.46 to 5.49 and degrading F_1 from 47.71% to 21.46%.
- Pure Sigmoid Focal Loss with single-cell assignment (1×1) achieves optimal convergence, reaching the highest AP₇₅ (38.98%) and AP_{50:95} (37.23%).

Table 5: Ablation of target assignment and loss formulations on the P_2 head.

Configuration	Target Assign	Loss Formulation	AP ₅₀	AP ₇₅	AP _{50:95}	UF-Rec	F ₁ @0.25	FPPI
Baseline BCE	1 × 1 Center	Standard BCE	62.58%	38.36%	36.99%	94.07%	47.71%	1.4667
Quality Focal Loss	3 × 3 Window	QFL ($\beta = 2.0$)	57.55%	37.01%	34.89%	93.01%	21.46%	5.4933
Pure Focal Loss [Ours]	1 × 1 Center	Focal ($\alpha = 0.25, \gamma = 2.0$)	62.13%	38.98%	37.23%	93.22%	46.78%	1.5278
All-in-One Combination	3 × 3 Window	QFL + Scale-Weights	57.55%	37.01%	34.89%	93.01%	21.46%	5.4933

6.3 Computational Efficiency & Edge Deployment Feasibility

In vehicular pavement inspection, real-time edge processing is mandatory. As summarized in Table 1 and Fig. 2, P2-DETR adds only 2.98M parameters to RT-DETR-L. On an NVIDIA RTX 4090 GPU, total pipeline latency is 47.4 ms (Backbone: 43.2 ms, P_2 Head + ScalePart NMS: 4.24 ms), delivering an operational throughput of **21.1 FPS** at 2K resolution. This easily satisfies the 20 FPS threshold for real-time mobile inspection vans traveling at 60 km/h.

7 Conclusion

In this work, we addressed the fundamental challenge of detecting ultra-fine potholes in 4K road imagery. Through comprehensive multi-paradigm benchmarking on HRP4K, we exposed the *4K Illusion* (brute-force scaling degrades accuracy and quadruples latency) and demonstrated the failure of spatial patch slicing (SAHI incurs multi-second delays and false alarm explosions). To resolve this paradox, we proposed **P2-DETR**, combining a frozen 2K Transformer detector with a lightweight dense P_2 auxiliary head ($stride = 4$, 2.98M parameters) trained via Sigmoid Focal Loss. By introducing Scale-Partitioned Prior NMS, we eliminated cross-scale interference, cutting post-processing latency to 4.24 ms and boosting Ultra-Fine Recall to **93.22%–94.07%**. Our work establishes a practical, real-time blueprint for high-resolution vehicular road distress inspection. Future work will explore learned cross-attention feature gating for end-to-end set prediction.

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